

So Let's Get Started

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Phase 1: Data Science Project Proposals

Project 1: Housing Amenities Impact on Apartment Listing Prices

Business Issue and Overview

Real estate investors and property managers lack quantitative evidence on which amenities drive rental pricing. Understanding amenity importance enables strategic property improvements and competitive pricing. This project analyzes which amenities (parking, laundry, pet policies, fitness centers, etc.) most significantly influence rental prices across U.S. markets.

Business Questions

- Which amenities have strongest impact on rental pricing?
- How does amenity importance vary across geographic regions?
- Which amenity upgrades yield the highest price premium relative to cost?
- Do certain amenity bundles (e.g., parking & laundry) have compounding effects on price?

Data Needs

Major entities: Rental listings (price, location, bedrooms, bathrooms, square footage), amenities (binary/categorical flags), geographic clusters (region, state), listing metadata (date listed, availability).

Reason

This project addresses a real business problem faced by property investors and managers daily. Understanding which amenities justify investment is directly actionable, it informs renovation priorities and pricing strategy. Personally, this project deepens familiarity with feature importance techniques and regression modeling, skills broadly applicable across industries.

Project 2: Student Course Success Prediction

Business Issue and Overview

Educational institutions face resource constraints and must identify at-risk students early. This project builds a predictive model to identify students likely to struggle or fail in courses, enabling early intervention programs. Predictions target first-semester students to maximize intervention effectiveness.

Business Questions

- Which student characteristics predict course success?
- When in the semester should interventions occur?
- What baseline GPA threshold indicates risk?
- How accurately can we identify at-risk students?

Data Needs

Major entities: Student demographics (age, prior GPA, major, course history), course characteristics (difficulty, subject, instructor), performance metrics (grades, attendance, engagement).

Reason

Addresses critical institutional challenge: student retention and success. Interest in educational technology and predictive analytics applied to real student outcomes. Classification modeling provides interpretable insights; understanding why students succeed enables actionable intervention strategies. Valuable for education stakeholders and demonstrates practical impact of data science on human outcomes.